

Cryptographic Tidbits

Lesson 3

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Cryptographic Tidbits

Lesson 3

- Machine Ciphers

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**INSERT
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- Machine Ciphers
- Automated Cryptanalysis

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- Machine Ciphers
- Automated Cryptanalysis
- Computational Security

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- Machine Ciphers
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- Computational Security
- Mathematical Security

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- Machine Ciphers
- Automated Cryptanalysis
- Computational Security
- Mathematical Security
- Enigma & Project Venona

Enigma

Cryptography on a new scale

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Computational Security

Quantifying the effort to break a cipher

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- Brute Force Attack (BFA) – try every possible key

Computational Security

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- Brute Force Attack (BFA) – try every possible key
- BFA requires $O(\text{key space})$ operations to break

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Computational Security

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Computational Security

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Computational Security

Quantifying the effort to break a cipher

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Computational Security

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- Computationally secure – best attack not much better than Brute Force

Computational Security

Quantifying the effort to break a cipher

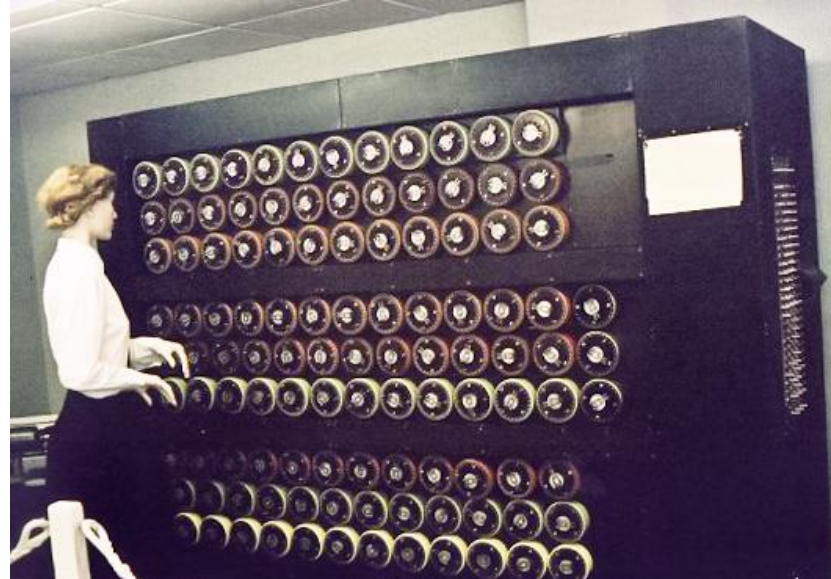
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- Computationally secure – best attack not much better than Brute Force
- Enigma – computationally insecure

Bletchley Park

Cryptanalysis on a new scale

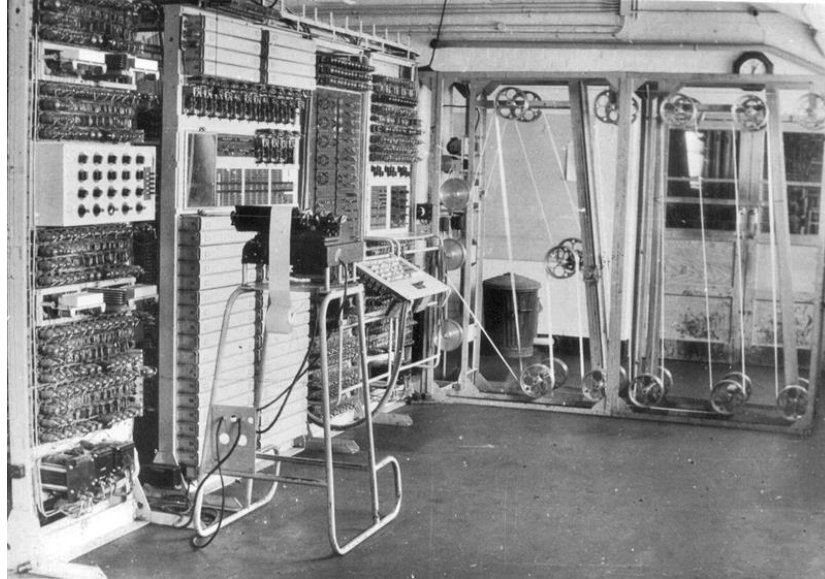
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Colossus

Fully-programmable digital computer

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The Vernam Cipher

Provably unbreakable

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Provably unbreakable

- Also known as One-Time Pad

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The Vernam Cipher

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- **Also known as One-Time Pad**
- **A page of random characters is the key**
- **Alice and Bob have identical copies the key**
- **Message/key combined one character at a time**
- **Every conceivable equally likely**
- **Each key page can only be used once**
- **Requires vast quantities of key material**

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- Requires vast quantities of key material
- Soviets reused key material in WWII

Venona

Breaking the provably unbreakable

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Cryptographic Tidbits

Lesson 3 in Review

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- Machine Ciphers represented quantum leap

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- Machine Ciphers represented quantum leap
- Computational Security

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- **Machine Ciphers represented quantum leap**
- **Computational Security**
- **One-Time Pad provably unbreakable**

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- **Machine Ciphers represented quantum leap**
- **Computational Security**
- **One-Time Pad provably unbreakable**
- **Cryptanalysis kept pace**